

COLUMBUS GLENN, OH, USA

WMO: 724280

Lat: 39.991N

Lon: 82.877W

Elev: 816

StdP: 14.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	9.8	-4.6	4.4	6.7	0.1	5.7	11.4	26.9	32.6	24.7	32.2	7.4	260	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	91.1	73.4	88.9	72.5	86.6	71.3	76.5	86.5	75.1	84.3	73.8	82.3	10.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	128.5	80.8	72.2	123.0	79.8	71.0	117.9	78.7	40.3	86.6	39.1	84.7	37.8	82.3	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.9	17.8	-0.9	94.2	7.3	2.7	-6.1	96.2	-10.4	97.8	-14.5	99.3	-19.8	101.3
			-1.8	79.0	7.0	1.7	-6.8	80.2	-11.0	81.2	-14.9	82.1	-20.0	83.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	29.4	32.7	41.8	53.7	63.5	72.3	75.4	74.3	67.7	55.8	43.9	34.8
	DBStd	18.52	12.38	11.33	11.00	9.26	8.31	6.10	4.93	4.83	6.96	8.73	9.37	10.06
	HDD50	2268	648	495	305	67	4	0	0	0	0	38	229	482
	HDD65	5161	1104	906	722	353	132	13	0	2	50	309	633	937
	CDD50	3683	9	9	50	177	424	668	786	753	532	218	47	10
	CDD65	1098	0	0	2	13	87	231	321	290	131	23	0	0
	CDH74	9470	0	1	20	132	769	1987	2884	2410	1081	184	2	0
	CDH80	3140	0	0	2	15	206	689	1052	801	336	39	0	0
Wind	WSAvg	7.8	9.4	9.2	9.2	9.2	7.6	7.0	6.2	5.9	6.0	7.1	8.1	8.6
Precipitation	PrecAvg	39.2	2.6	2.2	3.3	3.5	3.9	4.1	4.4	3.7	3.0	2.4	3.0	3.0
	PrecMax	56.4	9.0	5.2	10.2	7.2	9.8	10.4	12.4	11.5	7.3	6.7	10.7	7.0
	PrecMin	24.5	0.7	0.3	0.7	0.7	0.9	0.6	1.1	0.7	0.4	0.1	0.4	0.9
	PrecStd	6.7	1.4	1.1	1.7	1.5	1.8	2.0	2.1	2.0	1.7	1.3	1.7	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	67.4	76.4	82.3	88.4	92.9	94.8	92.9	91.8	84.7	71.4	64.2
		MCWB	57.8	58.3	60.6	64.5	70.3	73.4	76.6	73.9	71.8	68.8	60.3	59.0
	2%	DB	58.3	59.8	70.3	78.0	85.1	89.6	91.1	89.7	87.7	78.9	67.0	58.6
		MCWB	54.7	53.2	57.2	61.3	67.8	72.3	74.6	73.2	70.5	65.5	58.1	54.2
	5%	DB	53.2	54.8	64.6	73.8	82.2	86.9	88.6	87.2	83.9	74.6	62.9	54.4
		MCWB	49.8	49.0	54.4	59.4	67.2	70.9	73.1	72.4	69.2	62.8	55.6	50.7
	10%	DB	47.2	49.2	59.6	69.6	78.6	84.2	85.9	84.6	80.3	70.5	59.1	49.8
		MCWB	43.0	43.7	51.5	56.9	65.6	69.9	71.8	71.0	67.3	61.7	52.6	45.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	60.0	62.8	67.5	72.7	76.4	79.2	78.1	75.3	71.2	63.4	60.0
		MCDB	62.6	66.5	74.0	78.5	84.2	87.8	89.7	87.8	85.2	80.9	68.6	63.6
	2%	WB	55.4	54.6	59.2	64.2	70.7	74.7	77.1	76.1	73.3	68.4	59.9	55.3
		MCDB	58.4	58.4	66.9	73.4	81.0	85.4	87.7	85.1	82.3	75.2	65.1	57.6
	5%	WB	50.3	49.6	56.3	61.7	68.9	73.2	75.4	74.8	71.6	65.5	56.8	51.1
		MCDB	53.0	54.7	62.5	69.5	78.3	83.2	84.5	83.1	80.1	72.6	61.7	53.9
	10%	WB	43.7	44.0	52.5	59.3	67.3	71.8	74.1	73.2	69.9	62.4	53.3	45.9
		MCDB	46.9	48.9	59.7	67.2	75.9	81.0	82.2	80.7	77.2	69.3	58.4	49.3
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.1	17.4	20.2	19.3	18.4	18.0	18.1	19.3	18.6	15.9	12.8
		MCDBR	17.2	22.7	24.7	24.9	23.0	21.5	20.3	20.5	22.8	23.6	21.7	18.2
		MCWBR	14.8	17.8	15.0	13.9	10.2	8.7	7.4	7.7	9.1	11.9	15.0	15.6
	5% WB	MCDBR	16.6	20.9	20.9	20.7	19.3	18.5	18.0	17.3	18.5	19.4	18.4	17.3
MCWBR		15.1	18.1	14.6	13.4	9.6	8.5	7.7	7.8	8.3	10.9	14.6	16.3	
Clear-Sky Solar Irradiance	taub	0.310	0.330	0.360	0.397	0.438	0.456	0.465	0.456	0.397	0.365	0.328	0.306	
	taud	2.422	2.345	2.336	2.254	2.175	2.168	2.154	2.162	2.305	2.387	2.452	2.456	
	Ebn at Noon	271	280	283	278	268	262	259	258	267	264	261	263	
	Edh at Noon	26	32	36	42	47	47	47	46	37	31	25	23	
All-Sky Solar Radiation	RadAvg	517	785	1105	1496	1738	1905	1858	1718	1401	967	652	440	
	RadStd	49	102	102	132	144	133	106	89	127	86	87	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+121
(47)	Regional (42 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+121

Nomenclature: See separate page